Patient safety and hardware protection

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ISMRM Virtual Meeting: Vendor-Agnostic Pulse Sequence Programming with Pulseq II

Patient safety

- Peripheral (and cardiac) nerve stimulation
- RF power deposition (SAR)
- Acoustic noise

Hardware protection

- RF subsystem protection
- Gradient subsystem protection
- Avoiding mechanical resonances

Patient safety

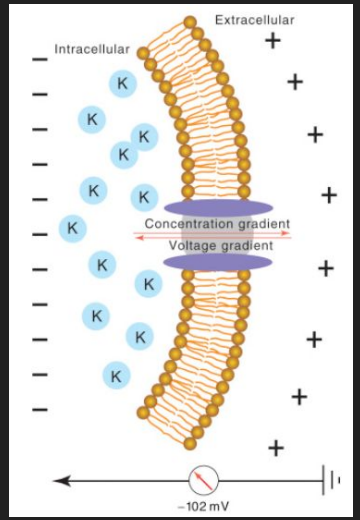
- Peripheral (and cardiac) nerve stimulation ✅
- RF power deposition (SAR) ✅
- Acoustic noise ❌

Hardware protection

- RF subsystem protection ✅
- Gradient subsystem protection ✅
- Avoiding mechanical resonances ✅

Peripheral nerve stimulation

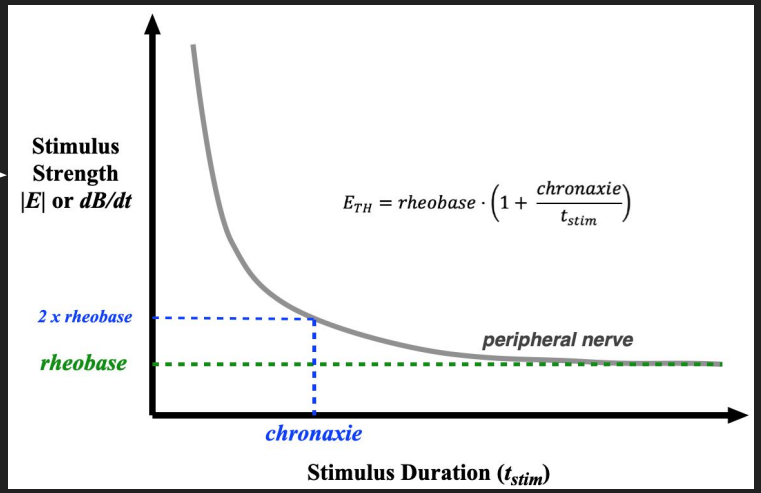
- dB/dt → dE/dt → transmembrane potential changes → nerve depolarization reaches threshold → action potential
- Direction dependent (L/R: 0.8; A/P: 1.0; S/I: 0.7)
- Goal: Predict PNS for a given Pulseq sequence
- Solution:
  - Model the PNS as a function of
    - Gradient slew rate waveform s(t)
    - biological response (charge buildup across membrane)
  - Vendor determines model parameters experimentally using test subjects' subjective PNS reporting (available on your scanner)



https://www.sciencedirect.com/topics/medicine-and-dentistry/membrane-steady-potential

Nerve impulse response model (GE)

- IEC 60601-2-33:2022
- Stimulation threshold model



- Resulting slew rate waveform:

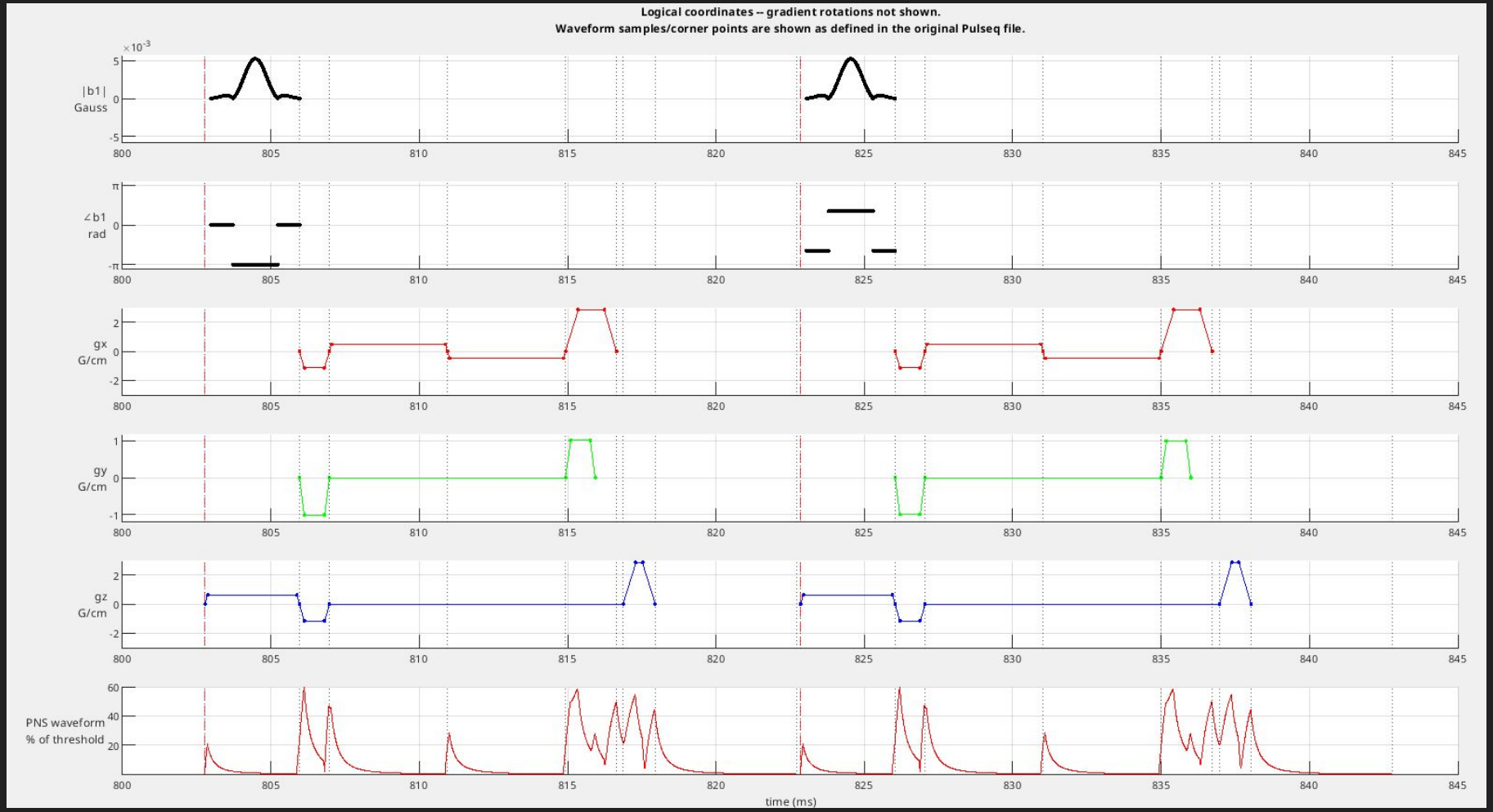
$$R(t) = \frac{1}{rb}\int_0^t \frac{\dot{B}(\theta)\, t_c}{(t_c + t - \theta)^2}\, d\theta$$

- `pge2.pns()` performs this calculation

+pge2

https://github.com/HarmonizedMRI/pge2/

pge2.plot()


Logical coordinates -- gradient rotations not shown.
Waveform samples/corner points are shown as defined in the original Pulseq file.
×10^-3
|b1|
Gauss
∠b1
rad
gx
G/cm
gy
G/cm
gz
G/cm
PNS waveform
% of threshold
time (ms)

PNS: what you need to do

1. Obtain PNS model parameters from your scanner
2. Run PNS calculation

```
%% GE
psq = pulseg.fromSeq('gre.seq');  % Load sequence

pge2.plot(psq, ...);    % or pge2.pns(psg, ...)


%% Siemens
seq.read('gre.seq');

seq.calcPNS(...);
```

RF power deposition (SAR)

- RF heats tissue via:
  - Resistive heating (induced currents in conductive tissues)
  - Dielectric heating (rotation of polar molecules like water)
- SAR (Specific Absorption Rate): Rate at which RF energy is absorbed by body tissue.
  - IEC 6-minute Average limit: 4.0 W/kg (body) / 3.2 W/kg (head)
  - IEC 10-second Average limit: Twice the 6-min average
- Affected by patient size, MRI sequence, field strength, coil design.

Two SAR checks: software and hardware

- Software check (interpreter)
  - Interpreter calls the vendor API safety functions (proprietary)
  - Underlying model (known only to manufacturer) factors in patient weight, RF pulse parameters, field strength, etc.
  - GE: API functions operate on each segment ⇒ interpreter performs a sliding window calculation
- Hardware monitor
  - Continuously measures power sent to coils (forward RF power) and reflected RF power (power not absorbed by the patient)
  - Produces a SAR <u>estimate</u> (not a direct measurement)
  - Always on, for Pulseq and native sequences alike
- You have no control over either of these (unless you modify the interpreter code) ⇒ There's nothing you need to do

RF subsystem protection

- Peak RF power
- RF duty cycle and power (heating)
- Interpreter performs software checks
  - checks peak b1 against the hardware limit
  - calls the (proprietary) vendor API functions to perform hardware safety checks
    - Underlying model is known only to the manufacturer
    - GE: per-segment calculation ⇒ sliding window calculation
- There’s nothing you need to do

Gradient subsystem protection

- Gradient amplifier heating (semiconductor devices); …
- Checks follow a similar workflow as the RF subsystem checks
- Interpreter calls vendor API (underlying model only known to manufacturer)
- GE:
  - per-segment calculation ⇒ sliding window calculation
  - Simulator (WTools) heating calculation typically matches the scanner
- There’s nothing you need to do

Avoiding mechanical resonances

- Forbidden frequencies
  - Scanner specific; provided to you by the manufacturer (config file on scanner computer)
- It is your responsibility to design EPI trains accordingly
  - Also: pCASL labeling trains; …
- Recommended: Plot sequence frequency spectrum (Siemens)

```
>> seq.gradSpectrum('path/to/asc_file_name');
```

- Do we need integrated/automatic checks?
  - Save EPI echo spacing in .seq file?
  - Should we require the interpreters to detect this?

Acoustic noise

- Oscillating gradients → Lorentz forces → mechanical vibrations & noise
- Knowledge gap in the research community (including me!)
- “Forbidden frequencies” address this -- but only partially?
- No universally accepted workflow exists ⇒ unmet need
  - But see presentation by Amir Seginer and Rita Schmidt, Pulseq user group meeting Dec 5, 2025
- Recommendation: work with your vendor…

Shutterstock.com

Key takeaways and recommendations

Your responsibilities:

- Check RF/gradient limits against scanner specs
- Obtain PNS model parameters from your scanner and run PNS check
- Check EPI/pCASL spacing (forbidden frequencies)

Recommendations

- Simulate sequence using the vendor simulator (IDEA; WTools; etc)
  - e.g., performs gradient heating calculations (GE)
- Consult with your vendor representative before installing and running demanding sequences (e.g., diffusion)
- Some systems may require extra caution (e.g., acoustic modeling)

Safety checks overview

	Scan designer (you)	Interpreter	Hardware monitor	Open/proprietary
PNS	✔	(✔)	(✔)	Open
SAR	(✔)	✔	✔	Proprietary
Acoustic noise	(✔)			
RF hardware		✔	?	Proprietary
Gradient hardware		✔	?	Proprietary
Mechanical resonances	✔			